

Errata for *Conceptual Replication of Four Key Findings about Factual Corrections and Misinformation during the 2020 US Election*

Fifteen corrections to *British Journal of Political Science* article 10.1017/S0007123422000631. None of them changes a conclusion of the article, though the fourth moves four published percentages by a material amount and the fifth is in the reference list. Every corrected value below is computed at render time from a full reproduction of the published analysis, held in the repository `active-maintenance-aec/coppock_etal_2023`. Percentages are given to one decimal place throughout, including in sentences where the article gave a whole number, so that every quantity in a corrected sentence is one the reproduction has verified. Bold marks the values that changed.

1. The reported total sample size

To study fact-checking during the 2020 election season, we conducted eight preregistered panel experiments (total **N = 17,681**).

Corrected: To study fact-checking during the 2020 election season, we conducted eight preregistered panel experiments (total **N = 17,629**).

The article’s own appendix Table 1 reports the eight panel sizes as 2,321, 2,090, 2,059, 2,390, 2,582, 2,200, 1,599 and 2,388, which sum to 17,629, and the deposited respondent data hold exactly that many.

2. The count of experiments in which misinformation reduced accuracy

Overall, exposure to the misinformation significantly decreased accuracy in **twelve** out of the twenty-four opportunities.

Corrected: Overall, exposure to the misinformation significantly decreased accuracy in **thirteen** out of the twenty-four opportunities.

Appendix Table 1 prints all thirteen in its unadjusted rows, and the deposited estimates and the reproduction agree on every one of them.

3. Two appendix floats carry the same number

Table 1: Sample composition by panel.

Table 1: Average treatment effects at Wave 1 among all subjects.

Corrected: the sample composition table on appendix page 3 and the Wave 1 average treatment effect table on appendix page 15 are two different floats, and both are captioned **Table 1**. The regression tables that follow are numbered 1 through 7 and the two tables of appendix section 7 are numbered 8 and 9, so the appendix’s table numbering runs 1, 1, 2, 3, 4, 5, 6, 7, 8, 9. A reader citing “appendix Table 1” cannot be understood.

4. The persistence percentages in the main text were not recomputed when the analysis was revised

The share of the Wave 1 correction effect that survives to a later wave is the quantity the pre-analysis plan defines and the appendix prints in the **ratio** row of each persistence table: the per-experiment ratio of the later-wave coefficient to the Wave 1 coefficient, pooled across the experiments by meta-analysis. The article reports that quantity in four sentences of the main text and again in the appendix, and the two disagree. Appendix Tables 5 and 7 print 0.49 and 0.43. The main text’s 66.4 and 50 are not returned by any pooling, adjustment or sample the deposited archive supports.

Both figures appear verbatim in versions of this manuscript that predate the analysis the article reports, which pooled 23 experiments rather than 24 and whose every cell differs from the published ones. They are the earlier analysis’s numbers, carried forward and never recomputed when the models were revised, rather than the product of a different estimator applied to the final data. A reader of the introduction therefore takes away a materially different number from a reader of the appendix.

The four sentences are corrected below to the random-effects pool of the covariate-adjusted ratios computed from the analysis the article actually reports, which is the estimator the appendix’s own meta-analysis rows use and the specification Figure 4 plots. Random effects should have been used throughout.

We show that **66 per cent** of the initial effect of fact-checks on factual beliefs is still detectable one week after initial exposure, with **50 percent** detectable after more than two weeks.

Corrected: We show that **48.9 per cent** of the initial effect of fact-checks on factual beliefs is still detectable one week after initial exposure, with **34.3 percent** detectable after more than two weeks.

The effects of fact-checks observed immediately after treatment dissipate somewhat from Wave 1 to Wave 2, on average, at **66.4 per cent** of the initial magnitude.

Corrected: The effects of fact-checks observed immediately after treatment dissipate somewhat from Wave 1 to Wave 2, on average, at **48.9 per cent** of the initial magnitude.

On average, the effect in Wave 3 is **50 per cent** the magnitude of the original effect, though the smaller number of issues and subjects increases our uncertainty.

Corrected: On average, the effect in Wave 3 is **34.3 per cent** the magnitude of the original effect, though the smaller number of issues and subjects increases our uncertainty.

The effects of fact-checks persisted at **66 per cent** the original magnitude after one week and at **50 per cent** after more than two weeks.

Corrected: The effects of fact-checks persisted at **48.9 per cent** the original magnitude after one week and at **34.3 per cent** after more than two weeks.

The Wave 3 correction is larger than the change of pooling alone would make it. The deposited three-wave models fitted with covariates regress the wrong outcomes, so the ratio behind appendix Table 7's 0.43 instruments the Wave 2 outcome rather than the Wave 3 outcome. Refitted on Wave 3, the random-effects pool is 34.3 per cent with covariates and 38.9 per cent without them.

One caveat the correction does not remove. A random-effects pool is still a precision-weighted average: it weights each fact-check by how precisely that fact-check's ratio was measured, not by how much of the population of fact-checks it stands for. The corrected percentages are therefore still not quite the quantity the sentences describe, which is how much of a typical fact-check's effect survives a week. They are the figures the article should have reported, and they are the ones a reader can check against the appendix; treating them as an estimate of the population average asks more of them than a precision-weighted average can give.

The reference list

The entry below corrects the article's reference list. It was found by an audit that sends every printed entry whole to Crossref and checks the authoritative record back into it, and it is verified against the record named in the note. No cited work is missing and no citation in the text points at the wrong work.

5. Rathje et al. (2022) is printed with one author

Published, p. 1341:

Rathje S (2022) Letter to the Editors of Psychological Science: Meta-analysis Reveals that Accuracy Nudges Have Little to No Effect for U.S. Conservatives: Regarding Pennycook et al.

Corrected:

Rathje S et al. (2022) Letter to the Editors of Psychological Science: Meta-analysis Reveals that Accuracy Nudges Have Little to No Effect for U.S. Conservatives: Regarding Pennycook et al.

Crossref, doi 10.31234/osf.io/945na: Rathje, Roozenbeek, Traberg, Van Bavel and van der Linden. Four coauthors are missing from the entry, which the reference list's own style would give as "Rathje S et al." The in-text citation on p. 1330 reads "Rathje et al. 2022", so the body of the article has it right.

6. Pennycook and Rand (2019) is printed with the wrong initial

Published, p. 1340:

Pennycook D and Rand DG (2019) Lazy, not biased

Corrected:

Pennycook G and Rand DG (2019) Lazy, not biased

Crossref, doi 10.1016/j.cognition.2018.06.011: Gordon Pennycook and David G. Rand. The same author is given correctly as “Pennycook G” in the 2021 Nature entry two lines below, and footnote 2 of the article cites “Pennycook and Rand 2019”, so only the initial in this entry is wrong. It appears to have been copied from Rand’s.

7. Do Facts Matter? is cited with its authors in the wrong order

Published, pp. 1328 and 1340:

Belief in falsehoods may shape political attitudes and behavior in ways that damage democratic accountability (**Einstein and Hochschild** 2015; Nyhan 2020)

Einstein KL and Hochschild J (2015) Do Facts Matter? Information and Misinformation in American Politics. Norman, OK: University of Oklahoma Press.

Corrected:

Belief in falsehoods may shape political attitudes and behavior in ways that damage democratic accountability ([Hochschild and Einstein]] 2015; Nyhan 2020)

[Hochschild JL and Einstein KL]] (2015) Do Facts Matter? Information and Misinformation in American Politics. Norman, OK: University of Oklahoma Press.

Library of Congress, LCCN 2014027612: “Do facts matter? : information and misinformation in American politics / Jennifer L. Hochschild and Katherine Levine Einstein”, Norman: University of Oklahoma Press, 2015. Hochschild is the book’s first author. The publisher and the city are as the entry gives them. The entry keeps the place the published list gave it, under E, because moving an entry is a re-presentation of the list rather than a correction to it; a reader looking for it will find it where the article filed it.

8. Swire-Thompson is cited under her earlier surname in a 2020 entry

Published, pp. 1331 and 1341:

The use of multiple outcome measures helps mitigate concerns about measurement error (**Swire, DeGutis, and Lazer** 2020)

Swire B, DeGutis J and Lazer D (2020) Searching for the Backfire Effect

Corrected:

The use of multiple outcome measures helps mitigate concerns about measurement error
([Swire-Thompson, DeGutis, and Lazer] 2020)

[Swire-Thompson B, DeGutis J and Lazer D] (2020) Searching for the Backfire Effect

Crossref, doi 10.31234/osf.io/ba2kc, and the published version at doi 10.1016/j.jarmac.2020.06.006, both give Briony Swire-Thompson. The 2017 Royal Society Open Science entry above it is correct as “Swire B”: the name changed between the two papers. As with entry 7, the entry keeps its published place in the list.

9. Porter and Wood (2022) is given the title of its preprint

Published, p. 1341:

Porter E and Wood TJ (2022) **Misinformation on the Facebook news feed: experimental evidence**. Journal of Politics 84, 1812-1817.

Corrected:

Porter E and Wood TJ (2022) Political misinformation and factual corrections on the Facebook news feed: experimental evidence. Journal of Politics 84, 1812-1817.

Crossref, doi 10.1086/719271: “Political Misinformation and Factual Corrections on the Facebook News Feed: Experimental Evidence”, The Journal of Politics 84(3), 1812-1817. The volume and the page range are the version of record’s; the title is the OSF preprint’s, which the entry also links. The two are different documents and the entry mixes them.

10. Nyhan (2020) is given a month where its issue number belongs

Published, p. 1340:

Nyhan B (2020) Facts and myths about misperceptions. Journal of Economic Perspectives **34(August)**, 220-236.

Corrected:

Nyhan B (2020) Facts and myths about misperceptions. Journal of Economic Perspectives 34(3), 220-236.

Crossref, doi 10.1257/jep.34.3.220: Journal of Economic Perspectives 34(3), 220-236. The doi the entry itself prints encodes the issue as 3. Every other entry in the list that gives an issue gives a numeral.

11. Dillard is dated a year before the article it cites appeared

Published, pp. 1328 and 1339:

culminated in an insurrection that policymakers have attributed to misinformation about the “big lie” of election fraud (Dillard **2020**; US House Committee on Energy and Commerce Staff, 2021)

Dillard J (**2020**) Ocasio-Cortez Reveals Past Sexual Assault, Faults GOP over Riot. Bloomberg.

Corrected:

culminated in an insurrection that policymakers have attributed to misinformation about the “big lie” of election fraud (Dillard [[2021]]; US House Committee on Energy and Commerce Staff, 2021)

Dillard J ([[2021]]) Ocasio-Cortez Reveals Past Sexual Assault, Faults GOP over Riot. Bloomberg.

The Bloomberg article is dated 2 February 2021, which the entry’s own URL records in its path, and it reports the aftermath of the 6 January 2021 attack on the Capitol. Its author is Jarrell Dillard, so the initial is right. Dated 2020 the entry describes a report published before the event it reports.

12. Shin and Thorson (2017) carries a capital left over from its title case

Published, p. 1341:

Shin J and Thorson K (2017) Partisan selective sharing: the biased diffusion of fact-checking messages on **social Media**. *Journal of Communication* 67(2), 233-255.

Corrected:

Shin J and Thorson K (2017) Partisan selective sharing: the biased diffusion of fact-checking messages on social media. *Journal of Communication* 67(2), 233-255.

The published title is “Partisan Selective Sharing: The Biased Diffusion of Fact-Checking Messages on Social Media” (doi 10.1111/jcom.12284). The reference list sets titles in sentence case, and the M of Media is the only capital left standing from the original.

Found while preparing the remastered edition

The entries below were found by reading every word of the article and its appendix against the published PDF while a corrected re-typesetting was being prepared, which is a finer sieve than any pass this repository had run. The last two describe properties of the appendix’s regression tables that the reproduction measures rather than values in a single cell.

13. A sentence break in the appendix has lost its space

Figures 4, 5, 6 display the coefficient **estimates. These** regressions return p-values below 0.05

Corrected: Figures 4, 5, 6 display the coefficient estimates. These regressions return p-values below 0.05

Found by reading the appendix against its own PDF while preparing the remastered edition. The two sentences are run together with no space after the full stop.

14. Confidence intervals in the appendix regression tables are printed low bound last

12.97 (2.01) **[16.91, 9.03**]

Corrected: the seven regression tables print **144 of their 492 intervals with the bounds transposed**, so that the interval reads from its upper limit to its lower one. The cell quoted above, the first in appendix table 1, should read [9.03, 16.91]. Every interval whose estimate is negated by the table code is affected and no other is.

The deposited script negates an estimate and both of its interval bounds without reversing them, so a two-sided interval comes out backwards whenever the sign is flipped. The bounds themselves are right and the reproduction returns the same two numbers for every affected cell; only their order on the page is wrong. The tables in the remastered edition are reproduced as published rather than reset, because the pipeline's own version of these tables orders and labels its rows differently as well, and re-presenting them would be a larger change than the one this entry describes.

15. Significance markers in the appendix regression tables were set by comparing strings

205 of the 492 cells in the seven regression tables carry a significance marker.

Corrected: **256** of them have a p-value below 0.05. The marker is on the wrong cells: "0.0144" sorts before "0.05" and earns a star, while "1.28e-10" sorts after it and does not, so the most significant estimates in these tables are among the unstarred ones.

The deposited table script reshapes each fit with a function whose identifier columns omit a character column, which coerces every value in the reshaped frame to a string; the surviving comparison with 0.05 is then a comparison of two strings rather than of two numbers. It is deterministic and reproducible: running the deposit's own script prints the published tables verbatim, star pattern included. As with entry 14, the tables in the remastered edition are reproduced as published rather than reset.